

Set 6: Product Sales Analysis

Dataset: Monthly Product Sales

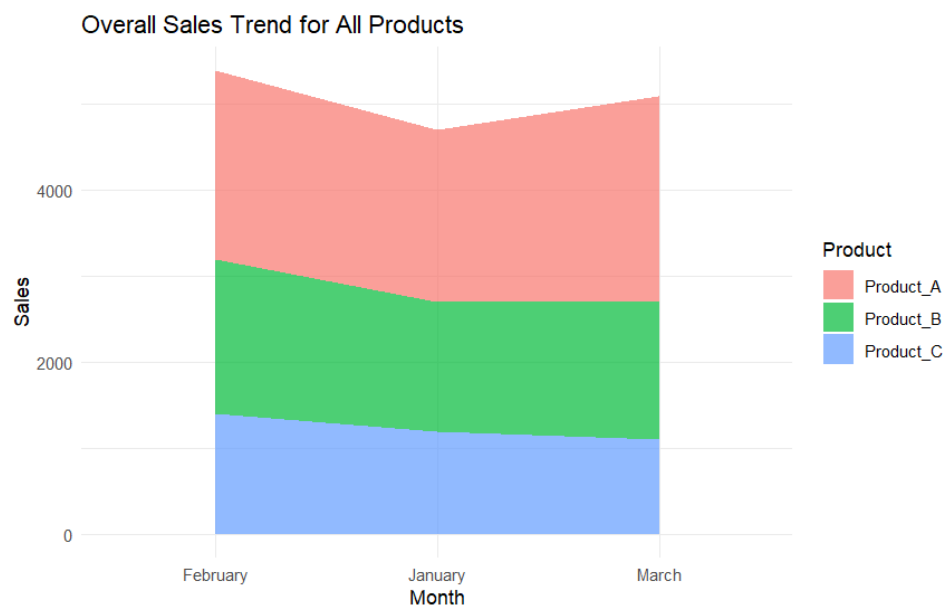
Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

```
1 sales <- matrix(c(2000, 1500, 1200,  
2                   2200, 1800, 1400,  
3                   2400, 1600, 1100),  
4                   nrow = 3,  
5                   byrow = FALSE)  
6 rownames(sales) <- c("Product A", "Product B", "Product C")  
7 colnames(sales) <- c("January", "February", "March")  
8 barplot(sales,  
9         beside = TRUE,  
10        col = c("skyblue", "lightgreen", "pink"),  
11        main = "First Quarter Sales by Product",  
12        xlab = "Month",  
13        ylab = "Sales",  
14        legend.text = rownames(sales),  
15        args.legend = list(title = "Products", x = "topright"))
```

First Quarter Sales by Product



```
1 install.packages("ggplot2")  
2 install.packages("reshape2")  
3 library(ggplot2)  
4 library(reshape2)  
5 sales <- data.frame(  
6   Month = c("January", "February", "March"),  
7   Product_A = c(2000, 2200, 2400),  
8   Product_B = c(1500, 1800, 1600),  
9   Product_C = c(1200, 1400, 1100)  
10 )  
11 sales_long <- melt(sales, id.vars = "Month",  
12                   variable.name = "Product",  
13                   value.name = "Sales")
```



```
1 sales_data <- data.frame(  
2   Product = c("Product A", "Product B", "Product C"),  
3   January = c(2000, 1500, 1200),  
4   February = c(2200, 1800, 1400),  
5   March = c(2400, 1600, 1100)  
6 )  
7 print(sales_data)|  
8 cat("Monthly Sales Data for Each Product\n\n")  
9 print(sales_data)
```

```
> print(sales_data)  
  Product January February March  
1 Product A    2000     2200  2400  
2 Product B    1500     1800  1600  
3 Product C    1200     1400  1100  
> |
```